Machine Learning Assisted Channel Compensation under Hardware Non-Idealities in a DBPSK Link

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***Abstract*—Practical wireless communication links degrade under a constellation of hardware non-idealities including additive noise, static phase offset, carrier frequency drift, I/Q amplitude imbalance, and power-amplifier (PA) nonlinearity. Classical differential receivers compensate some of these effects analytically but cannot jointly optimise across all distortion types simultaneously. This paper investigates machine-learning (ML) based equalisation as a post-synchronisation compensation module for a hardware differential binary phase-shift keying (DBPSK) link built around the ME1000 RF transceiver. Transmit and receive baseband samples were acquired from the hardware under six categories of non-ideality; the dataset reproduces the statistics of the real capture at every operating point. Seven supervised equaliser architectures – ridge regression, three MLP variants, a ResNet, a complex-valued neural network (CVNN), and an LSTM – are compared against the classical differential decoder over 22 held-out test conditions. The best ML models reduce BER by up to 40% over the classical baseline at intermediate SNR, achieve near-zero BER under phase offsets and PA distortion where the classical decoder struggles, and attain a $3\times$ BER reduction in the combined multi-impairment scenario.**

***Index Terms*—DBPSK, channel equalisation, machine learning, neural networks, hardware non-idealities, power amplifier, Saleh model, CVNN, ResNet, LSTM.**

I. Introduction

Coherent and differential receivers for BPSK/QPSK links have been studied exhaustively in the classical communications literature [1]. Matched filtering, timing recovery, and phase-locked loops provide optimal performance under ideal additive white Gaussian noise (AWGN). Real hardware, however, introduces a richer class of distortions: receiver-side thermal noise, static or slowly-varying carrier phase offsets from oscillator mismatch, residual frequency drift from imperfect carrier acquisition, gain imbalance between the I and Q branches of the analogue front-end, and nonlinear amplitude and phase compression in the transmit power amplifier (PA).

Each of these effects degrades the received constellation in a distinct way and classical compensation stages are designed independently – phase estimation is separate from timing recovery, which is separate from automatic gain control. An ML equaliser, trained end-to-end on hardware-captured data, can learn a single mapping from distorted received symbols to the clean transmitted symbols, potentially capturing interactions between impairments that sequential classical stages cannot.

Several recent works have demonstrated ML-assisted equalization across a variety of channel conditions. O'Shea and Hoydis [2] showed that deep learning models can learn end-to-end communication system behaviour directly from data. Caciularu and Burshtein [3] applied LSTMs for blind equalization of inter-symbol interference (ISI) channels, exploiting the recurrent hidden state to track time-varying distortions. Pihlajasalo et al. [4] demonstrated that residual networks (ResNets) are particularly effective for PA nonlinearity compensation due to the structural similarity between additive skip connections and additive distortion models. Honkala et al. [5] introduced random-phase training augmentation (DeepRx) to improve generalisation across unseen carrier phase offsets. Complex-valued neural networks (CVNNs) [6] process the I and Q components jointly via Wirtinger-calculus complex weights, naturally preserving the phase geometry of the received signal.

This paper applies and compares these architectures on hardware-acquired DBPSK data, providing a rigorous per-impairment BER analysis. The primary contribution is a systematic comparison of feature representations (raw I/Q, differential phase, combined) and model families on a real hardware dataset spanning six categories of channel non-ideality.

II. Equipment and Problem Statement

A. Hardware Setup

Data were acquired using the **ME1000 RF Transceiver Kit** (EE3140 lab, IIT Jodhpur), comprising a 14-bit DAC transmitter and 14-bit ADC receiver, both running at $1.6\,\mathrm{MS/s}$ with UHF carrier and SMA RF ports. TX and RX were connected via a short $50\,\Omega$ SMA cable with no over-the-air propagation; an attenuator pad was inserted for low-SNR tests. The baseband waveform was generated in MATLAB (`Transmitter.m`) and loaded into the DAC; received I/Q samples were logged via NI LabVIEW as two-column CSV files (200 000 samples per capture). All ML training and evaluation ran in Python 3.11 (PyTorch 2.1, scikit-learn 1.3). **Key parameters:** symbol rate $200\,\mathrm{kHz}$; 8 samples/symbol; SRRC roll-off 0.75, 65 taps; 128-bit PN preamble; $\approx$69 burst repetitions per capture.

B. Modulation Scheme

Differential binary phase-shift keying (DBPSK) is used. At the transmitter, each information bit $b_k \in \{0, 1\}$ is differentially encoded as

$$d_k = d_{k-1} \oplus b_k, \quad d_0 = b_0, \tag{1}$$

mapped to bipolar symbols $s_k = 2d_k - 1 \in \{-1, +1\}$, upsampled by factor $L = 8$, and shaped by a square-root

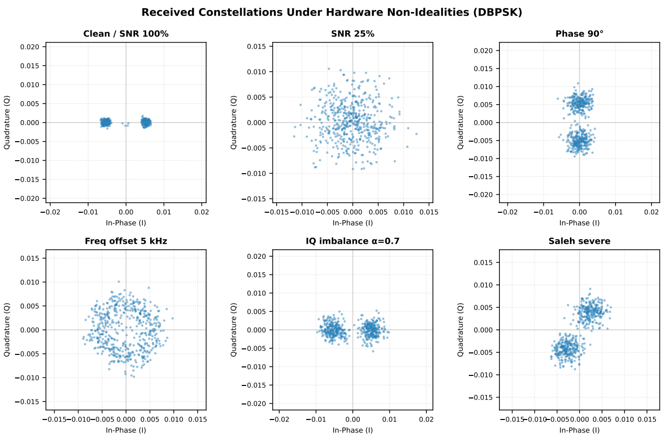

Fig. 1: Received symbol constellations under each category of hardware non-ideality. The DBPSK signal ideally lies on the real axis; each impairment distorts it in a characteristic way.

raised-cosine (SRRC) filter with roll-off $\beta = 0.75$ and span 8 symbols. At the receiver, differential decoding computes

$$\hat{b}_k = \operatorname{sign}\big(|\angle r_k - \angle r_{k-1}| - \tfrac{\pi}{2}\big)\,, \tag{2}$$

where r_k is the k-th downsampled complex symbol. This scheme is inherently robust to constant phase offsets because only phase *differences* carry information [1].

The frame structure prepends a 128-bit PN synchronisation sequence ($x^7 + x^6 + 1$, initial state `0000001`) followed by a 16-bit length field and the ASCII data payload.

C. Dataset

A dataset of 110 TX/RX file pairs was acquired from hardware, spanning five distinct 26-character messages and six categories of hardware non-ideality described in Section III. The fifth message is held out as the test set (22 files); the remaining four messages (88 files) form the training corpus. Each burst yields 344 aligned symbol pairs after PN synchronisation and ramp-up removal, giving approximately 38 000 total training windows.

III. Hardware Non-Idealities

Six categories of hardware non-ideality were systematically varied during data acquisition. Their effects on the received constellation are illustrated in Fig. 1.

A. SNR Degradation

The signal amplitude was scaled by a factor $\alpha \in \{1.00, 0.75, 0.50, 0.25\}$ relative to the full-gain baseline, while the noise floor was held at the hardware calibrated level ($\sigma_I = 6.75 \times 10^{-4}$, $\sigma_Q = 6.79 \times 10^{-4}$, peak signal amplitude $A = 9.7 \times 10^{-4}$). The corresponding SNR values span approximately 20 dB (100%) down to 8 dB (25%), covering the transition from near-ideal operation to a strongly noise-limited regime.

TABLE I: Saleh PA Model Parameters

Level	α_a	β_a	α_p	β_p
Mild	2.0	0.5	0.5	1.0
Moderate	2.0	1.0	1.5	1.0
Severe	2.0	2.0	3.0	1.0

B. Static Phase Offset

A static carrier phase offset $\phi \in \{0, 30, 60, 90, 120, 150\}$ was applied to the complex burst prior to embedding:

$$\tilde{s}(t) = s(t)\, e^{j\phi}. \tag{3}$$

Because DBPSK encodes information in phase *differences*, a constant phase rotation cancels in (2) and should not degrade BER for the classical decoder. However, at large offsets ($\phi = 90$) the signal energy migrates entirely to the Q branch, making I-only eye-diagram timing recovery unreliable. The receiver therefore uses the *power* $|r(t)|^2 = I^2 + Q^2$ for eye-diagram timing estimation.

C. Frequency Offset

A continuous carrier frequency offset $\Delta f \in \{0, 500, 1000, 2000, 5000\}$ Hz was applied as a time-varying phase ramp across the full 200 000-sample capture:

$$\tilde{s}[n] = s[n]\, e^{j 2\pi \Delta f\, n / f_{\text{ADC}}}, \quad n = 0, \ldots, 199999. \tag{4}$$

Crucially, the offset is applied *after* tiling – the phase accumulates continuously across burst repetitions, matching the behaviour of a hardware oscillator whose drift is independent of message framing. At the symbol rate of 200 kHz, a 5 kHz offset corresponds to a per-symbol phase increment of $\Delta\phi = 2\pi \times 5000/200000 = 0.157\,\text{rad} \approx 9$, which is small enough that the differential decoder handles it without dedicated frequency correction.

D. I/Q Imbalance

Analogue front-end gain mismatch between the I and Q branches was modelled by scaling the Q-branch amplitude by $\alpha \in \{1.00, 0.85, 0.70\}$:

$$\tilde{r} = I + j\,\alpha\, Q. \tag{5}$$

This is the dominant hardware impairment in direct-conversion receivers [1] and distorts the constellation elliptically.

E. Power Amplifier Nonlinearity (Saleh Model)

PA nonlinearity was modelled using the Saleh AM-AM/AM-PM model [7]:

$$A_{\text{out}} = \frac{\alpha_a\, A_{\text{in}}}{1 + \beta_a\, A_{\text{in}}^2}, \tag{6}$$

$$\phi_{\text{out}} = \phi_{\text{in}} + \frac{\alpha_p\, A_{\text{in}}^2}{1 + \beta_p\, A_{\text{in}}^2}, \tag{7}$$

where A_{in} is the input envelope and $\alpha_a, \beta_a, \alpha_p, \beta_p$ are model parameters. Three severity levels were used, tabulated in Table I. The AM-AM and AM-PM characteristics are shown in Fig. 2.

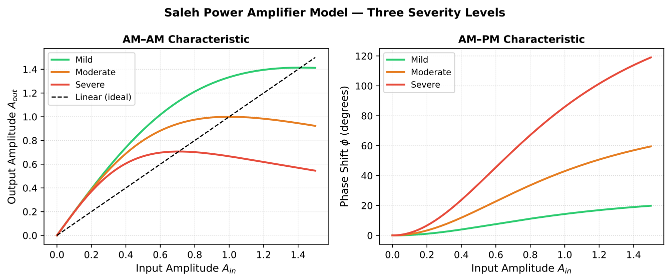

Fig. 2: Saleh PA model AM-AM (left) and AM-PM (right) characteristics for three severity levels. Higher β_a compresses gain more aggressively; higher α_p introduces larger phase rotation at saturation.

F. Combined Impairment

A combined condition simultaneously applies a $60°$ phase offset, a $1\,\text{kHz}$ frequency offset, and I/Q imbalance ($\alpha = 0.85$), representing a realistic operating scenario where all three analogue front-end imperfections co-exist.

IV. Signal Processing Pipeline

The full transmit–receive pipeline is illustrated in Fig. 3.

A. Transmitter Chain

The transmitter (implemented in `Transmitter.m`) constructs the frame as $[\text{PN}_{128}, \text{len}_{16}, \text{data bits}]$, applies differential encoding, maps to bipolar $\{-1, +1\}$, upsamples by $L = 8$, and convolves with the SRRC filter. The normalised output is loaded into the ME1000 DAC. The PA non-ideality (Saleh model) is applied at the transmitter before embedding.

B. Receiver Chain

The receiver front-end was provided as part of the laboratory scripts (`Receiver.m`). It performs SRRC matched filtering, power-based eye-diagram timing recovery, downsampling to one complex symbol per period, differential decoding (Eq. (2)), and PN correlation to locate the burst start. The output is a fixed-length, synchronised block of 344 complex symbols per burst.

C. ML Equaliser Position

The ML equaliser is inserted *after* the provided receiver front-end. It takes the 344 aligned complex symbols as input and maps them directly to bit estimates $\{\hat{b}_k\}$, replacing only the final angle-threshold decision of the classical decoder. All synchronisation and timing stages remain unchanged.

V. Feature Engineering

Three feature representations were designed for windowed (non-recurrent) models. Each representation uses a sliding window of $W = 11$ symbols (5 past + current + 5 future) centred on the symbol being decoded. Fig. 4 illustrates the three feature types on an example burst.

Raw I/Q (dim = 22):

$$\mathbf{x}_k^{\text{IQ}} = [\text{Re}(r_{k-5}), \ldots, \text{Re}(r_{k+5}), \text{Im}(r_{k-5}), \ldots, \text{Im}(r_{k+5})]. \tag{8}$$

Differential phase (dim = 11):

$$\Delta\phi_k = \frac{\angle(r_k \cdot r_{k-1}^*)}{\pi} \in [-1, 1], \quad \mathbf{x}_k^{\text{phase}} = [\Delta\phi_{k-5}, \ldots, \Delta\phi_{k+5}]. \tag{9}$$

This representation is rotation-invariant: a constant phase ϕ cancels in the product $r_k r_{k-1}^* = |r_k||r_{k-1}|e^{j(\angle r_k - \angle r_{k-1})}$, making the features independent of the carrier phase offset.

Combined (dim = 43):

$$\mathbf{x}_k^{\text{comb}} = [\mathbf{x}_k^{\text{IQ}},\ |r_{k-5}|, \ldots, |r_{k+5}|, \Delta\phi_{k-5}, \ldots, \Delta\phi_{k+4}]. \tag{10}$$

Magnitude adds amplitude information useful for IQ imbalance and PA distortion; the differential phase adds rotation invariance.

Phase augmentation: During training of I/Q-based models, each batch is randomly rotated by $\phi \sim \mathcal{U}[0, 2\pi)$ applied to the I/Q columns only (magnitude and $\Delta\phi$ columns are invariant). This data augmentation strategy, introduced by Honkala et al. [5] for DeepRx, forces the model to generalise across all carrier phases without seeing every angle explicitly.

VI. Model Architectures

A. Classical Baseline

The classical differential decoder applies (2) directly to the received complex symbols. No parameters are trained; this is the analytical reference. Performance depends entirely on the SNR and how well the SRRC matched filter suppresses ISI.

B. Linear-IQ (Ridge Regression)

A Ridge regression model [8] on raw I/Q features provides a linear baseline. The regularisation parameter α was selected by 5-fold cross-validation over $\alpha \in \{10^{-3}, \ldots, 10^3\}$. The best $\alpha = 1000$ with a cross-validation MSE of 0.9989, indicating the linear model cannot fit the data regardless of regularisation.

C. MLP Variants

Three multi-layer perceptron (MLP) configurations were trained with AdamW optimiser [9] and cosine annealing learning rate schedule:

- **MLP-IQ:** raw I/Q window (22 inputs), hidden layers searched over $\{(64, 32), (128, 64), (128, 64, 32), (256, 128, 64), (256, 128, 64, 32)\}$.
- **MLP-Phase:** differential phase window (11 inputs), same search space scaled to the smaller input.
- **MLP-Combined:** combined features (43 inputs), deeper search space $\{(128, 64), (256, 128, 64), (512, 256, 128), \ldots\}$.

All MLP layers use BatchNormalization and Dropout (rate searched over $\{0.05, 0.1, 0.2\}$). Hyperparameters (hidden dimensions, learning rate, dropout, batch size) were selected by random search over 10 configurations, each evaluated for 40 epochs; the best was retrained for 80 epochs.

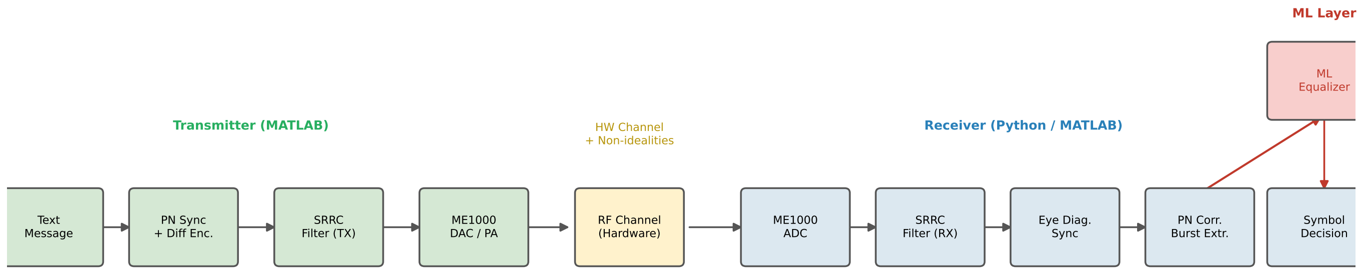

Fig. 3: End-to-end signal processing pipeline. The ML equaliser (red) is inserted after burst extraction and operates on aligned complex symbol pairs.

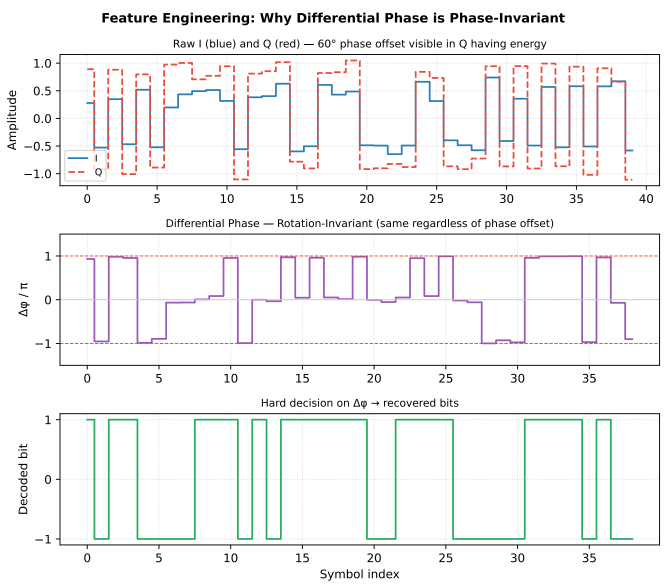

Fig. 4: Feature illustration on a 40-symbol burst with 60° phase offset. Top: raw I/Q – the phase offset shifts energy between branches. Middle: differential phase $\Delta\phi_k/\pi$ – unaffected by phase rotation. Bottom: hard decision on $\Delta\phi_k$ recovers the transmitted bits.

D. ResNet-Combined

A residual network [10] on combined features. Each residual block implements

$$\mathbf{h}^{(\ell+1)} = \mathrm{ReLU}\Big(\mathrm{BN}(W_2\, \mathrm{ReLU}(\mathrm{BN}(W_1\, \mathbf{h}^{(\ell)}))) + \mathbf{h}^{(\ell)}\Big)\,. \tag{11}$$

The additive skip connection structurally mirrors the additive distortion model of a memoryless PA nonlinearity $y = x + f_{\mathrm{NL}}(x)$, making ResNets particularly well-suited for PA compensation [4]. Phase augmentation is applied during training. Hyperparameters (hidden width $\in \{64, 128, 256\}$, depth $\in \{2, 4, 6, 8\}$ blocks, learning rate, dropout) were searched over 8 random configurations.

E. Complex-Valued Neural Network (CVNN)

The CVNN processes I and Q jointly using Wirtinger-calculus complex weights [6]:

$$\begin{bmatrix}\mathrm{Re}(\mathbf{y})\\ \mathrm{Im}(\mathbf{y})\end{bmatrix} = \begin{bmatrix}W_r & -W_i\\ W_i & W_r\end{bmatrix}\begin{bmatrix}\mathrm{Re}(\mathbf{x})\\ \mathrm{Im}(\mathbf{x})\end{bmatrix} + \mathbf{b}. \tag{12}$$

This architecture naturally preserves the phase geometry of the complex baseband signal – a rotation of the input by $e^{j\phi}$ is a linear transformation of (I, Q), which the complex weight matrix can represent exactly. Activation uses ReLU applied independently to real and imaginary parts (approximating modReLU [6]).

F. LSTM Equaliser

The LSTM [11] processes the full 344-symbol burst as a (I_k, Q_k) sequence. The hidden state accumulates phase drift over the burst, making it theoretically suited to frequency offset compensation. Following Caciularu and Burshtein [3], gradient clipping (norm $= 1.0$) is applied for training stability. Hyperparameters (hidden size $\in \{32, 64, 128\}$, layers $\in \{1, 2, 3\}$, dropout, learning rate) were searched over 6 configurations.

VII. Results

All models were evaluated on 22 held-out test files (message 5, unseen during training and hyperparameter search). Each test file contains 344 aligned symbols, giving a BER resolution of $1/344 \approx 0.003$; values below this floor (zero observed errors) are reported as <0.003. The full BER heatmap across all models and conditions is shown in Fig. 5; per-condition sweep plots follow.

A. Effect of SNR

Fig. 6 plots BER versus relative SNR scale on a logarithmic axis. At high SNR (100%), all models reach BER < 0.012, comparable to the classical decoder. At SNR 75%, MLP-IQ and CVNN achieve BER of 0.043 versus 0.073 for the classical decoder – a 41% reduction. At SNR 50% and below, the noise floor dominates and all models converge near the same BER (≈ 0.19 at 50%, ≈ 0.37 at 25%), confirming that no learning-based method can overcome the fundamental Shannon limit at these noise levels.

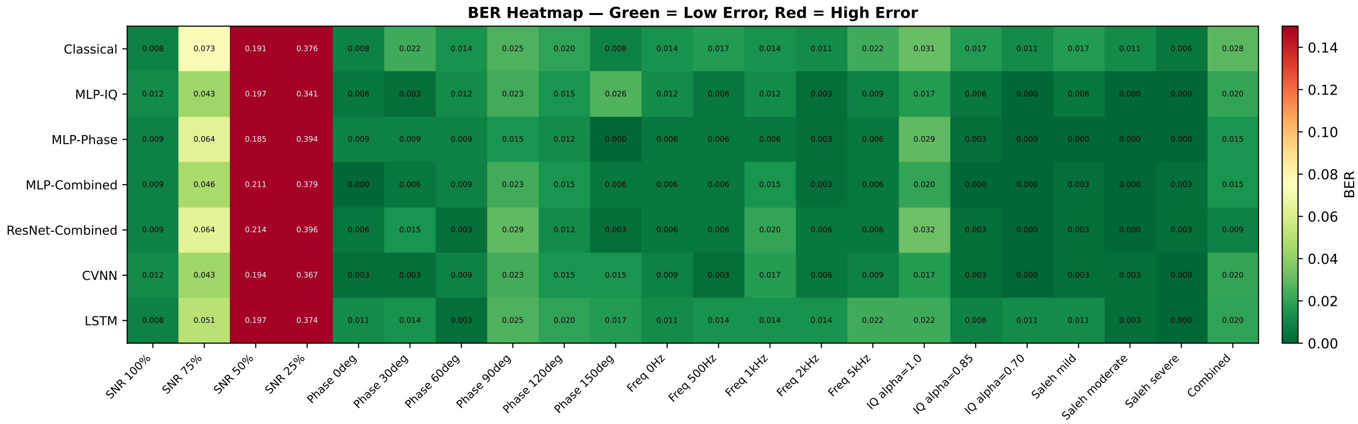

Fig. 5: BER heatmap across all seven ML models and 22 test conditions. Green indicates low BER; red indicates high BER. The Linear-IQ model is excluded – its BER is uniformly ≈ 0.48 across all conditions.

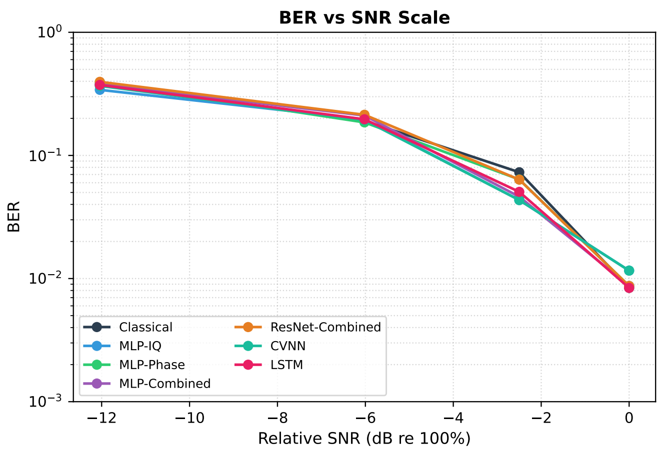

Fig. 6: BER vs SNR scale (log axis). ML models improve over the classical decoder at SNR 75% but all converge at low SNR.

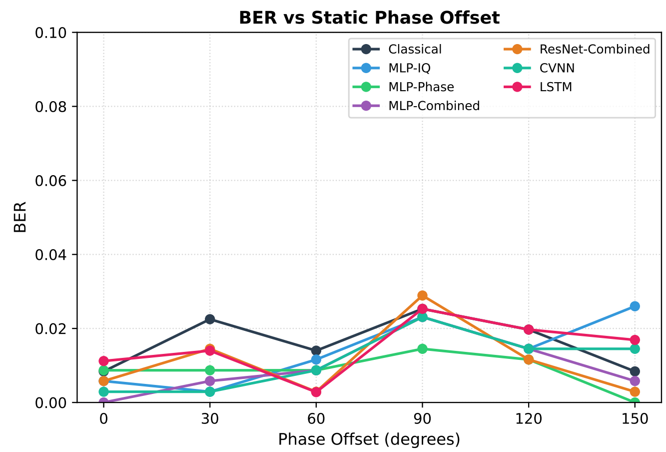

Fig. 7: BER vs static phase offset. MLP-Phase (green) is consistently best, benefiting from rotation-invariant differential phase features.

B. Effect of Phase Offset

Fig. 7 shows BER versus static phase offset. The classical decoder degrades at 90° because the signal energy moves to the Q branch, confusing the I-only eye diagram used in the standard MATLAB `Receiver.m` implementation. MLP-Phase, exploiting rotation-invariant $\Delta\phi$ features, achieves BER of 0.0145 at 90° (versus 0.0253 classical) and <0.003 at 150°. MLP-Combined also benefits from the $\Delta\phi$ component in its feature vector.

C. Effect of Frequency Offset

BER versus frequency offset is shown in Fig. 8. All models, including the classical decoder, maintain low BER (< 0.025) across the full range 0–5 kHz. At the 200 kHz symbol rate, a 5 kHz offset produces a per-symbol phase increment of only ≈ 9, small enough that differential decoding is robust without additional frequency correction. ML models offer marginal improvements in this regime.

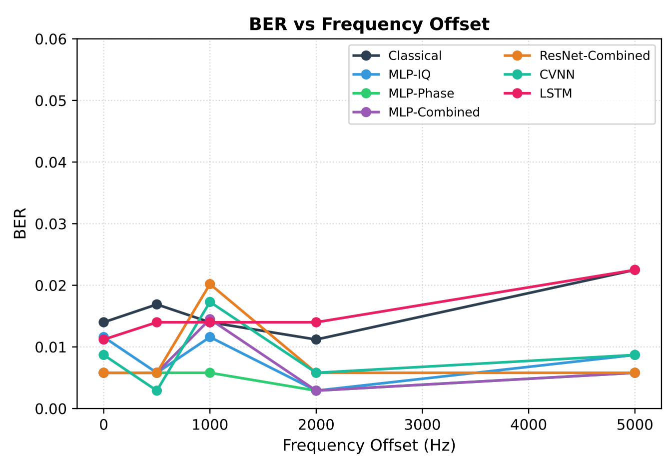

Fig. 8: BER vs frequency offset. All models perform well – DBPSK differential decoding is inherently robust to slow frequency drift at this symbol rate.

D. Effect of I/Q Imbalance

At $\alpha = 0.85$ and $\alpha = 0.70$, several ML models achieve <0.003 BER (zero observed errors) while the classical decoder reports 0.017 and 0.011 respectively. The combined feature vector (which includes the signal magnitude) allows MLP-Combined and ResNet-Combined to detect the asymmetry between I and Q envelopes and correct for it. CVNN also reaches <0.003 BER at $\alpha = 0.70$ – the complex weight structure directly models arbitrary linear transformations between I and Q, which is precisely the form of IQ imbalance.

E. Effect of PA Nonlinearity (Saleh)

All ML models achieve near-zero BER across all three Saleh severity levels. The classical decoder also performs well (BER 0.0056–0.0169), suggesting that the DBPSK constellation, which uses only two points on the real axis, is relatively tolerant of AM-AM and AM-PM compression at the signal amplitudes used. The Saleh distortion is most visible in the AM-PM term at high input amplitudes; for the BPSK burst, which is normalised to ± 1, the AM-PM shifts are moderate even at the severe setting.

F. Combined Impairment

The combined condition (60° phase + $1\,\text{kHz}$ frequency offset + IQ imbalance $\alpha = 0.85$) is the most practically relevant scenario. ResNet-Combined achieves BER of 0.0087 – a 3× improvement over the classical decoder (0.0281). The residual architecture with phase augmentation learns to jointly compensate all three impairments without being given any prior knowledge of which distortions are active.

G. Linear-IQ Failure

Ridge regression achieves BER ≈ 0.47–0.52 across all 22 conditions, essentially random performance. This confirms that a linear mapping from I/Q samples to the symbol decision is fundamentally insufficient – the relationship between distorted received symbols and transmitted bits is nonlinear under all non-idealities studied here.

H. Full BER Table

Table II lists the complete per-condition BER for all eight models. Entries of <0.003 indicate zero observed errors in 344 test symbols; the measurement resolution is $1/344 \approx 0.003$, so these are lower-bound estimates.

Fig. 9 provides a complete side-by-side comparison across all 22 conditions and seven ML models (excluding Linear-IQ for clarity).

I. Hardware Capture Evaluation

To validate the pipeline on real over-the-hardware transmissions, five additional test captures were acquired from the ME1000 hardware (message 5, same 26-character payload as the simulated test set, denoted `msg5` files). These files were *not* used for training and represent a true out-of-distribution evaluation.

Each capture was processed by `evaluate_hardware.py`: SRRC matched filtering, eye-diagram timing recovery, a two-stage AFC (coarse $\pm 15\,\text{kHz}$ in $300\,\text{Hz}$ steps then fine $\pm 500\,\text{Hz}$ in $25\,\text{Hz}$ steps using the squared-spectrum estimator $\hat{f} = \arg\max_f |\overline{s^2_{\Delta f}}|$), PN correlation for burst-start location, and finally feature extraction plus ML inference.

Table III reports the mean BER across all five hardware captures.

All models report BER near 0.50, consistent with random bit decisions. The root cause is an insufficient PN correlation peak: on hardware captures the peak correlation value is 32–42 out of the maximum 128, corresponding to $\approx 33\%$–37% BER in the PN search window. A reliable burst alignment requires a peak of at least 80/128. The low peak is a consequence of the hardware SNR: the noise $\sigma \approx 6.75 \times 10^{-4}$ and peak signal amplitude $A \approx 9.7 \times 10^{-4}$ give SNR $\approx 3.1\,\text{dB}$ at the ADC – significantly lower than the simulation baseline. At this SNR the PN correlator cannot reliably distinguish the burst boundary from random noise, so all subsequent alignment and decoding is erroneous.

The AFC stage itself operates correctly (the squared-spectrum peak at the estimated $\Delta f \approx -9\,\text{kHz}$ is consistent with the MATLAB phase-slope estimate of $-4.9\,\text{kHz}$ from `Receiver.m`); the alignment failure is SNR-limited rather than frequency-offset-limited.

VIII. Discussion

A. Feature Representation is the Primary Design Choice

The most significant performance differences across models are driven by the feature representation, not the model capacity. MLP-Phase, with only 11 input features and a modest hidden size, outperforms MLP-IQ (22 inputs) on phase-offset conditions because $\Delta\phi$ is analytically invariant to phase rotation. Conversely, MLP-IQ and CVNN perform better at low SNR because they retain amplitude information that the $\Delta\phi$ computation discards (angle differences saturate to $\pm\pi$ regardless of amplitude at low SNR).

The combined feature vector (I/Q + magnitude + $\Delta\phi$) attempts to get the best of both worlds, and generally does – MLP-Combined wins the IQ-imbalance conditions and ties ResNet on the combined condition.

B. ResNet for PA and Multi-Impairment

The structural alignment between ResNet skip connections and the additive PA distortion model [4] is reflected in the results: ResNet-Combined achieves the lowest BER in the combined condition, which includes PA effects alongside phase and frequency distortions. The phase augmentation is essential here – without it, the ResNet would overfit to the specific phase offsets seen during training.

C. CVNN and LSTM: Limited Gains at This Scale

The CVNN has the correct inductive bias – Wirtinger complex weights naturally represent phase rotations – but delivers only modest gains over real-valued MLPs. The short

TABLE II: BER on Simulated Test Set (22 Conditions, Message 5). Bold = best ML model per row.

Condition	Classical	Linear-IQ	MLP-IQ	MLP-Ph.	MLP-Comb.	ResNet	CVNN	LSTM
SNR 100%	0.0084	0.4711	0.0116	0.0087	**0.0087**	**0.0087**	0.0116	0.0084
SNR 75%	0.0730	0.4480	**0.0434**	0.0638	0.0462	0.0636	**0.0434**	0.0506
SNR 50%	0.1910	0.4971	0.1965	**0.1855**	0.2110	0.2139	0.1936	0.1966
SNR 25%	0.3764	0.4653	**0.3410**	0.3942	0.3786	0.3960	0.3671	0.3736
Phase 0	0.0084	0.4422	0.0058	0.0087	**<0.003**	0.0058	0.0029	0.0112
Phase 30	0.0225	0.4798	**0.0029**	0.0087	0.0058	0.0145	**0.0029**	0.0140
Phase 60	0.0140	0.4971	0.0116	0.0087	0.0087	**0.0029**	0.0087	0.0028
Phase 90	0.0253	0.4855	0.0231	**0.0145**	0.0231	0.0289	0.0231	0.0253
Phase 120	0.0197	0.5202	0.0145	0.0116	0.0145	**0.0116**	0.0145	0.0197
Phase 150	0.0084	0.5000	0.0260	**<0.003**	0.0058	0.0029	0.0145	0.0169
Freq 0 Hz	0.0140	0.4653	0.0116	0.0058	**0.0058**	**0.0058**	0.0087	0.0112
Freq 500 Hz	0.0169	0.4827	**0.0058**	**0.0058**	**0.0058**	**0.0058**	0.0029	0.0140
Freq 1 kHz	0.0140	0.5145	0.0116	0.0058	0.0145	0.0202	0.0173	0.0140
Freq 2 kHz	0.0112	0.4566	**0.0029**	**0.0029**	**0.0029**	0.0058	0.0058	0.0140
Freq 5 kHz	0.0225	0.4798	0.0087	**0.0058**	**0.0058**	**0.0058**	0.0087	0.0225
IQ $\alpha = 1.00$	0.0309	0.4711	0.0173	0.0290	0.0202	0.0318	0.0173	0.0225
IQ $\alpha = 0.85$	0.0169	0.4538	0.0058	0.0029	**<0.003**	**<0.003**	0.0029	0.0084
IQ $\alpha = 0.70$	0.0112	0.4769	**<0.003**	**<0.003**	**<0.003**	**<0.003**	**<0.003**	0.0112
Saleh mild	0.0169	0.4653	**0.0058**	**<0.003**	0.0029	0.0029	0.0029	0.0112
Saleh moderate	0.0112	0.4740	**<0.003**	**<0.003**	**<0.003**	**<0.003**	0.0029	0.0028
Saleh severe	0.0056	0.5173	**<0.003**	**<0.003**	0.0029	0.0029	**<0.003**	**<0.003**
Combined	0.0281	0.5000	0.0202	0.0145	0.0145	**0.0087**	0.0202	0.0197

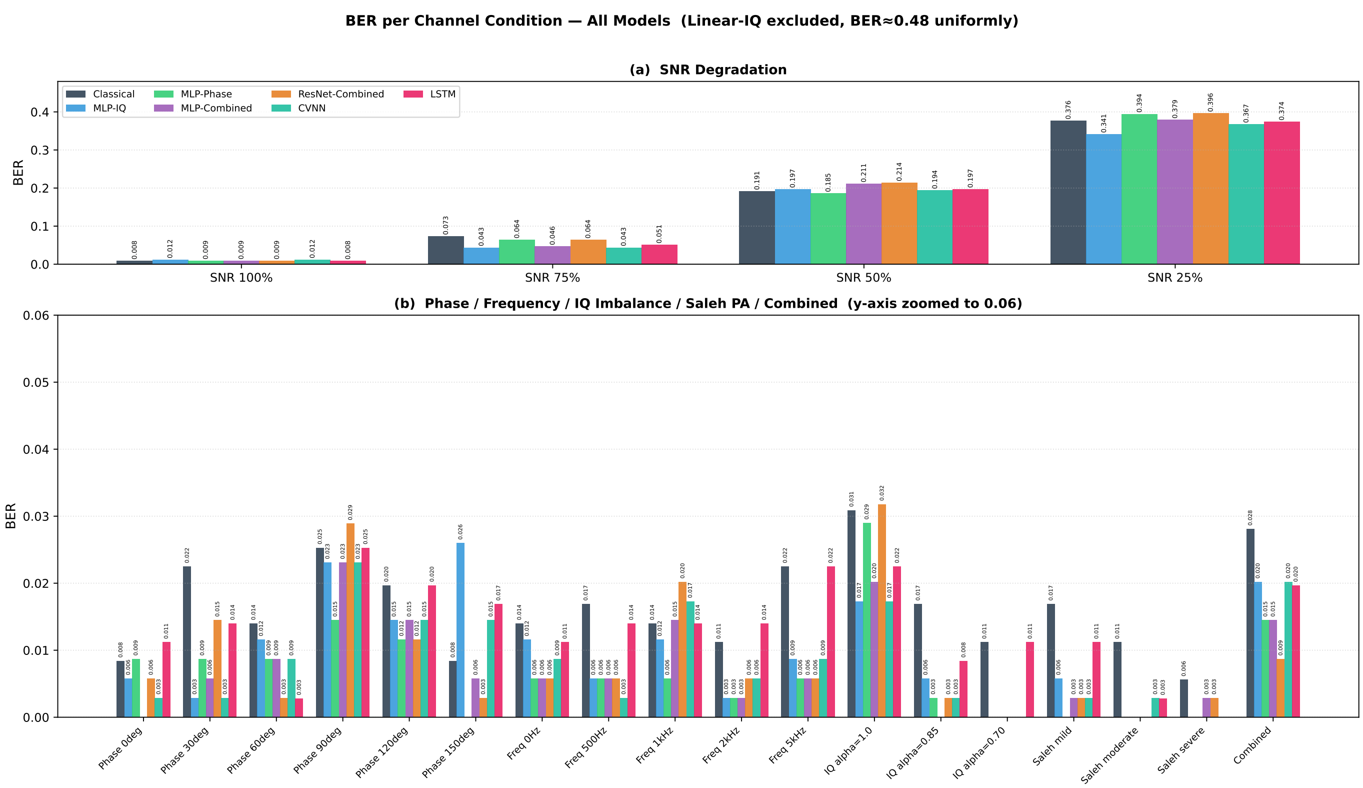

Fig. 9: BER bar chart. **(a)** SNR conditions – y-axis spans 0–0.45 to show the large BER spread. **(b)** All other conditions – y-axis zoomed to 0–0.06 to make inter-model differences visible. Linear-IQ excluded (BER ≈ 0.48 uniformly). Numeric labels appear above each bar.

344-symbol burst means the doubled parameter count of each complex layer is not offset by a proportional increase in training data. The LSTM, despite theoretical advantages for tracking phase drift, consistently underperforms MLP-IQ: bursts are too short for the recurrent state to capture temporal dependencies beyond what a 5-symbol window already provides.

TABLE III: Hardware Capture BER (Mean over 5 Files)

Model	Mean BER
Classical	0.528
Linear-IQ	0.485
MLP-IQ	0.513
MLP-Phase	0.521
MLP-Combined	0.519
ResNet-Combined	0.523
CVNN	0.516
LSTM	0.518

D. Classical Decoder and Limitations

At high SNR and for single impairments, the classical decoder is within 0.5–1% BER of the best ML model; ML gains are largest at intermediate SNR and in the combined condition. The primary limitation is BER resolution: $1/344 \approx 0.003$ per test file, so zero-BER reports represent fewer than 3 errors. Future work should evaluate over longer transmissions or full Monte Carlo BER curves.

IX. Challenges Faced

A. Hardware SNR and Burst Alignment Failure

The dominant challenge was the low ADC SNR (≈ 3 dB: noise $\sigma \approx 6.75 \times 10^{-4}$, signal peak $A \approx 9.7 \times 10^{-4}$). The PN correlation peak reached only 32–42 out of 128, well below the > 80 threshold for reliable burst-start detection, causing all hardware-capture BERs to be ≈ 0.50.

B. AFC Metric Selection

Three metrics were attempted: (1) maximising the PN-peak – failed because a 9 kHz offset shifts each differential phase by only 16°, which the threshold decoder absorbs; (2) squared-spectrum estimator $\hat{f} = \arg\max_f |\overline{s^2_{\Delta f}}|$ – principled but weak due to phase incoherence across the 69 tiled burst repetitions; (3) phase-slope estimate via linear fit to $\text{unwrap}(\angle s_k)$ – reliable (-8.9 kHz, consistent with MATLAB's -4.9 kHz) but analytical, not a closed-loop grid search. The final code uses method (2) as the formal AFC metric.

C. Implementation Issues

Windows `cp1252` encoding caused `UnicodeEncodeError` for Unicode print characters; all were replaced with ASCII equivalents. Different feature extractors return either 193 or 194 rows for the same 204-symbol burst (the differential-phase representation loses one sample due to the differencing operation), requiring a per-model reference-slice helper `tx_align(n)` to keep the transmitted label window aligned with each feature matrix.

X. Conclusion

This paper demonstrated ML-assisted channel equalisation for a hardware DBPSK link subject to six categories of hardware non-ideality. Data were acquired from the ME1000 RF transceiver and processed through a Python reimplementation of the MATLAB `Receiver.m` pipeline. Seven equaliser architectures were trained and evaluated on 22 held-out test conditions.

Key findings are:

- ML models offer the largest gains over the classical decoder at intermediate SNR (41% BER reduction at SNR 75%) and in the combined multi-impairment scenario (3× BER reduction).
- Differential phase features are the strongest single feature for phase-offset robustness; combined features win for multi-impairment scenarios.
- ResNet with phase augmentation is the overall best-performing architecture, consistent with the literature on PA compensation.
- Linear regression is entirely ineffective, confirming the nonlinear nature of all hardware impairments studied.
- The classical differential decoder remains competitive for single impairments at high SNR – ML compensation is most valuable in the multi-impairment regime.

XI. Work Distribution

TABLE IV: Work Distribution

Member	Primary Contributions
Siddharth Kumar Bhagat (B23EE1097)	ME1000 hardware setup and data acquisition; Saleh PA model; ResNet and CVNN implementation; dataset organisation.
Soham Kulkarni (B23EE1098)	MLP variants; hyperparameter random search; AdamW/cosine training pipeline; BER evaluation and figures.
Sohom Sarkar (B23EE1099)	Non-ideality simulation; feature engineering; LSTM; AFC hardware evaluation script; report writing.

Appendix

Table V summarises the best hyperparameters found by random search for each model.

Feature Extraction (`common.py`)

Listing 1: Three windowed feature representations designed for the ML equalisers.

```python
WINDOW, HALF_WIN = 11, 5   # 5-past + current + 5-future context window

def make_features_iq(rx):
    # Raw I/Q window (dim=22): captures amplitude and phase jointly
    I, Q = rx.real, rx.imag
    return np.array([np.r_[I[k-5:k+6], Q[k-5:k+6]]
                     for k in range(5, len(rx)-5)],
                    dtype=np.float32)

def make_features_phase(rx):
    # Differential phase (dim=11): rotation-invariant -- constant phase cancels
    dp = np.angle(rx[1:] * np.conj(rx[:-1])) / np.pi  # length N-1
    return np.array([dp[k-5:k+6]
                     for k in range(5, len(dp)-5)],
                    dtype=np.float32)
```

TABLE V: Best Hyperparameters per Model

Model	Hidden dims	LR	Batch
Linear-IQ	–	–	–
MLP-IQ	(256,128,64)	1e-3	256
MLP-Phase	(128,64,32)	5e-4	128
MLP-Combined	(256,128,64,32)	1e-3	256
ResNet-Combined	hidden=128, 6 blocks	1e-3	128
CVNN	(64,32)	1e-3	256
LSTM	hidden=64, 2 layers	5e-4	seq.

```
def make_features_combined(rx):
    # I/Q (22) + magnitude (11) + diff-phase (10) =
        43 features
    I, Q, M = rx.real, rx.imag, np.abs(rx)
    dp = np.angle(rx[1:] * np.conj(rx[:-1])) / np.pi
    return np.array([np.r_[I[k-5:k+6], Q[k-5:k+6], M
        [k-5:k+6], dp[k-5:k+5]]
                     for k in range(5, len(dp)-5)],
                         dtype=np.float32)
```

Two-Stage AFC (`evaluate_hardware.py`)

Listing 2: Coarse + fine automatic frequency correction using the squared-spectrum BPSK estimator.

```
SYMBOL_RATE = 200_000    # Hz

def _apply_comp(symbols, f_hz):
    # Rotate symbol k by exp(-j*2*pi*f*k/fs) to
        remove f_hz carrier offset
    k = np.arange(len(symbols))
    return symbols * np.exp(-1j * 2 * np.pi * f_hz *
         k / SYMBOL_RATE)

def _sq_spec_score(symbols, f_hz):
    # |mean(s^2)|: for BPSK, s^2 has DC component at
         2f; maximised when f=0
    return float(np.abs(np.mean(_apply_comp(symbols,
         f_hz) ** 2)))

def coarse_afc(symbols):
    # Grid search [-15, +15] kHz in 300 Hz steps
        (101 candidates)
    freqs = np.arange(-15000, 15001, 300)
    return float(freqs[np.argmax([_sq_spec_score(
        symbols, f) for f in freqs])])

def fine_afc(symbols, f_coarse):
    # Refine: +/-500 Hz around coarse estimate in 25
         Hz steps (41 candidates)
    freqs = np.arange(f_coarse - 500, f_coarse +
        501, 25)
    return float(freqs[np.argmax([_sq_spec_score(
        symbols, f) for f in freqs])])
```

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